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Research Article

Development and Validation of a Multiclass Model Defining Molecular Archetypes of Kidney Transplant Rejection: A Large Cohort Study of the Banff Human Organ Transplant Gene Expression Panel

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ABSTRACT

Gene expression profiling from formalin-fixed paraffin-embedded (FFPE) renal allograft biopsies is a promising approach for feasibly providing a molecular diagnosis of rejection. However, large-scale studies evaluating the performance of models using NanoString platform data to define molecular archetypes of rejection are lacking. We tested a diverse retrospective cohort of over 1400 FFPE biopsy specimens, rescored according to Banff 2019 criteria and representing 10 of 11 United Network of Organ Sharing regions, using the Banff Human Organ Transplant panel from NanoString and developed a multiclass model from the gene expression data to assign relative probabilities of 4 molecular archetypes: No Rejection, Antibody-Mediated Rejection, T Cell—Mediated Rejection, and Mixed Rejection. Using Least Absolute Shrinkage and Selection Operator regularized regression with 10-fold cross-validation fitted to 1050 biopsies in the discovery cohort and technically validated on an additional 345 biopsies, our model achieved overall accuracy of 85% in the discovery cohort and 80% in the validation cohort, with ≥75% positive predictive value for each class, except for the Mixed Rejection class in the validation cohort (positive predictive value, 53%). This study represents the technical validation of the first model built from a large and diverse sample of diagnostic FFPE biopsy specimens to define and classify molecular archetypes of histologically defined diagnoses as derived from Banff Human Organ Transplant panel gene expression profiling data.

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Introduction

Gene expression profiling (GEP) of kidney allograft tissue has been extensively investigated over the last few decades.[1-8] Tissue-based GEP provides an orthogonal view of intragraft disease, complementing histologic assessment and offering a wide range

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of continuous values, which could be useful for refining broad diagnoses, stratifying disease activity, or confirming diagnoses when histologic biomarkers are absent or incompletely sampled.[9,10] Recognizing the significance of GEP in the context of Antibody-Mediated Rejection (ABMR) of kidney allografts, the Banff Foundation for Allograft Pathology expanded diagnostic criteria in 2013 to include transcriptional data.[11] However, intragraft GEP remains infrequently used in clinical practice, with the only commercially available test being a microarray-based analysis of fresh tissue.[1,3,6,12,13] Adoption of the microarray approach has been hampered due to reluctance of clinicians to take additional tissue solely for molecular testing and to some degree by concern that sampling bias may lead to mismatches in molecular and histologic phenotypes.[12]

GEP studies of formalin-fixed paraffin-embedded (FFPE) renal allograft biopsies, the routine tissue format for existing diagnostic workflows, have shown promise as an alternative approach.[4,8,14-17] For example, the NanoString nCounter platform for GEP is tolerant of relatively low-quality fragmented RNA, such as that derived from FFPE tissue. This approach makes feasible not only large-scale retrospective studies using archived clinical samples but also prospective reflex or secondary assessment of biopsy samples taken during routine clinical practice. The Banff Foundation officially endorsed the NanoString nCounter platform and the Banff Human Organ Transplant (B-HOT) panel at their 2019 consensus conference.[9,18] The Banff Molecular Diagnostics Working Group developed the B-HOT gene panel following comprehensive review of published gene expression data from biopsies of multiple types of solid organ allografts to identify transcripts representative of the range of rejection and tissue injury phenotypes and relevant molecular pathways, as well as from internal reference genes recommended by NanoString Technologies.[9] Recently published studies performed either by GEP of FFPE renal allograft biopsies using the B-HOT gene panel on the nCounter platform,[19-22] or by *in silico* analysis of microarray gene expression data for genes included in B-HOT panel,[23,24] have strongly supported the value of data from the B-HOT panel for scientific investigation and its potential for clinical application. Therefore, the B-HOT panel, designed by consensus, informed by decades of literature on transcriptional profiles in human organ allografts, could be thought of as the base tool from which to develop the molecular counterpart to the Banff histologic consensus for evaluation of allografts.[25]

In this study, we aimed to develop and technically validate a supervised model based on GEP data from the B-HOT panel using a large cohort of FFPE biopsies to define molecular archetypes of rejection as correlated with well-defined histologic diagnoses. More than 1400 cases were identified and processed from the archives of Arkana Laboratories, a referral-based diagnostic renal pathology practice in the United States. Based on data from a discovery cohort of 1050 biopsies representing the major diagnostic categories of rejection and including a diverse range of histologic features of non-rejection diagnoses, we developed a multiclass model to define and classify molecular archetypes of ABMR, T Cell—Mediated Rejection (TCMR), and Mixed Rejection. This model was then technically validated in an independent cohort of 345 biopsies.

Materials and Methods

Cohort Selection

Cases were screened by review of Arkana Laboratories archives from January 1, 2012, to November 30, 2021, with additional single instances of rare diagnoses from 2008 to 2011. All biopsies had undergone routine diagnostic workup by a board-certified, fellowship-trained renal pathologist. All biopsies were processed at Arkana Laboratories, and material from all cases was examined by light microscopy, immunofluorescence microscopy, and electron microscopy, except that electron microscopy was not performed for <10% of samples in both discovery and validation cohorts due to an insufficient amount of viable cortical tissue for ultrastructural examination. The tissue for light and electron microscopy was formalin-fixed, and the tissue utilized for immunofluorescence studies was preserved in Michel's transport medium until processed.

Both media were provided by Arkana Laboratories in biopsy kits sent to the referring institutions). Immunofluorescence included IgA, IgG, IgM, C3, C1q, C4d, albumin, fibrinogen, and kappa and lambda light chain staining for all cases and was graded on a 0 to 3+ scale. Ultrastructural features recorded included the presence/absence of electron-dense deposits, presence/absence of subendothelial lucent widening and new glomerular basement formation, presence/absence of peritubular capillary basement membrane multilayering, as well as the degree of podocyte foot process effacement. All biopsies were stained for SV40 immunohistochemistry (CellMarque), C4d staining (BIO-RAD), and were scored by Banff 2019 criteria, including rereview and scoring of cases diagnosed prior to implementation of the Banff 2019 scoring criteria.[18,26] Screened cases were filtered for defined histologic subgroups, and all TCMR Banff grade 2A or 2B cases also had $i \geq 2$ and $t \geq 2$; all ABMR cases were $i \leq 1$ and $t \leq 1$, except that 3 ABMR cases in the discovery cohort and 1 ABMR case in the validation cohort were either t2 i1 or t1 i2. In ABMR cases where the donor-specific antibody (DSA) status was unknown at the time of biopsy, the patient's clinician was contacted for this information. Exclusion criteria included failure to meet minimal Banff tissue adequacy criteria (ie, <7 glomeruli or <2 arteries), diagnosis of >1 histologic class (eg, polyomavirus nephritis [PVN] with TCMR grade 2A), concomitant immune complex-mediated disease (except mild IgA deposition), unknown DSA status in histologic ABMR despite attempts to obtain this information, insufficient residual tissue. Materials from all screened cases were reviewed by 1 of 9 expert renal pathologists (C.C., C.B., N.M., T.C., L.C., S.S., J.A., C.L., S.C.) for agreement with the original diagnosis in all cases and scoring of inflammation in fibrotic interstitial areas of scarred cortex (i-IFTA; IFTA, interstitial fibrosis and tubular atrophy), tubulitis in scarred cortical areas (t-IFTA), and total cortical inflammation including both scarred and nonscarred cortex (ti) in cases prior to 2020, with consensus reached on any discrepancies. After model validation, cases were annotated with patient age and sex, sample age, and referral center state/nation. The study protocol was approved by Solutions Institutional Review Board and conformed to the Declaration of Helsinki principles.

RNA Extraction and Gene Expression Analysis by nCounter

Two tissue curls of each FFPE block were cut at 16-μm thickness using RNase-free technique on a Leica RM2135 microtome (Leica Biosystems) and stored at −80 °C until RNA extraction with Zymo Quick-RNA FFPE kits (Zymo Research) essentially as described by the manufacturer with only minor modifications to the steps of deparaffinization and de-crosslinking conditions (as described in the Supplementary Material). For 16.5% of biopsies, RNA was extracted using a Zymo Quick-RNA MagBead kit (Zymo Research) and Tecan Fluent Liquid Handler. RNA concentration was determined using Qubit RNA HS Assay Kit with Qubit 3

Fluorometer (Invitrogen) as instructed by the manufacturer. RNA was assayed by the nCounter B-HOT panel (NanoString) essentially as described by the manufacturer with a target input of 150 ng total RNA (in 8 μL) in each hybridization reaction. When RNA yield was <150 ng, all obtained RNA was used as input for hybridization. Samples with low RNA concentration were vacuum-concentrated before hybridization.

Gene Expression Raw Data Processing and Quality Control

Raw counts of gene expression data from nCounter analysis were subject to normalization using 12 housekeeping genes prior to further analysis. The standard normalization was performed by the manufacturer's recommended procedure, except that housekeeping gene normalization was performed not per cartridge but after pooling all samples in the discovery cohort. Raw count gene expression data for each sample in the discovery and validation cohorts were normalized by housekeeping genes to the entire sample pool of all 1050 samples in the final discovery cohort prior to further analysis. The B-HOT gene panel assays/reagents used were of 2 different lots. To mitigate possible batch effect on raw gene expression data due to lot change, the ComBat-seq procedure was used to process the raw gene expression data.[27] Quality control (QC) metrics were based on NanoString nCounter data analysis guidelines and published literature and included housekeeping gene scaling factor (HKG SF) <10, and >62% probes above the detection threshold.[28] Samples failing to meet QC metrics were excluded from further analysis.

Diagnostic Model Fitting

Model fitting was performed using the caret package in R.[29] The goal of our model was to assign a molecular archetype/class to each biopsy representing 1 of 4 histologic diagnostic groups/classes (no rejection, ABMR, TCMR, or mixed rejection) based on the GEP of the sample as assayed by the B-HOT gene panel. After excluding 13 genes whose expression is specific to either heart, liver, or lung by B-HOT gene panel annotations,[9] the remaining 745 B-HOT panel genes were included in the candidate gene pool for model fitting. The final model is the result of a "Least Absolute Shrinkage and Selection Operator" (LASSO) fit to the set of eligible samples in the discovery cohort by 10-fold cross-validation to determine the 1se lambda value – the value of the lambda hyperparameter for the most parsimonious model with a cross-validated misclassification rate within 1 standard deviation of the minimum cross-validated misclassification rate.[30,31] For each biopsy sample, the multiclass classification model generates 4 different probability scores, with each score corresponding to 1 of the 4 molecular classes. The mathematic values of the 4 scores sum to 1. The molecular diagnosis is the class with the highest probability score. More detailed information on model development and the choice of LASSO over other potential models is available in the Supplementary Material.

Additional Data Analysis and Statistical Methods

Additional information on data analysis and statistical methods, including differential gene expression (DGE) analysis, is available in the Supplementary Material.

Results

Construction of Final Discovery and Validation Cohorts

Biopsies representing defined histologic subgroups were assigned to 1 of 4 histologic classes as follows: ABMR, either active or chronic active (A_H); Acute TCMR of various grades (T_H); Mixed Rejection, defined as meeting Banff histologic criteria for both ABMR and TCMR (M_H); or No Rejection, composed of various histologic subgroups failing to meet diagnostic criteria for rejection, (NV_H). Notably, the NV_H group included kidney allograft biopsies as well as native kidney biopsies, either lacking inflammation or with inflammation attributed to viral or bacterial infection. A total of 1085 biopsy specimens were assigned to the discovery cohort, with an additional 350 biopsy specimens assigned to the validation cohort, aiming for a balanced proportion of histologic classes between cohorts (Fig. 1). The time from biopsy procedure to RNA extraction (sample age) ranged from 85 to 4724 days (mean 1125, SD 861) in the discovery cohort, and from 28 to 4428 days (mean 837, SD 833) in the validation cohort. Samples failing QC metrics comprised 35 samples in the discovery

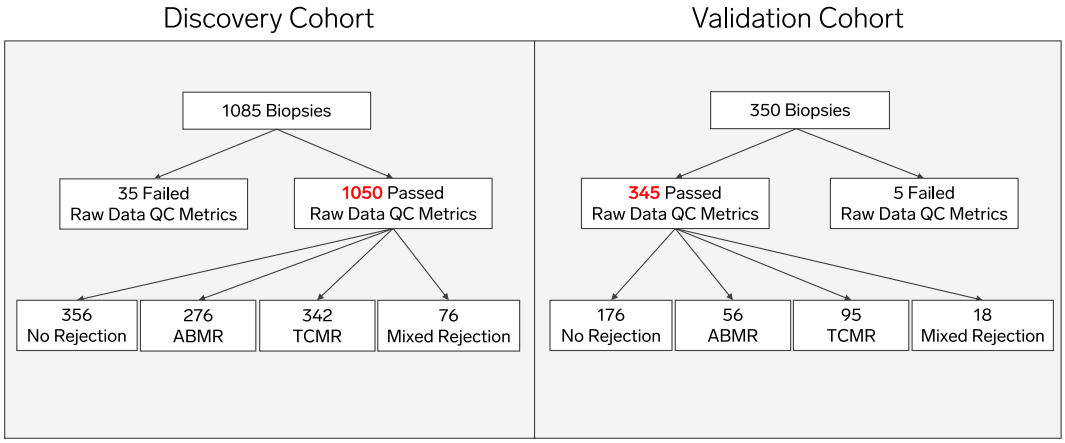

Figure 1.
Sample cohorts for discovery and validation. Left panel: Discovery cohort, n = 1085 initial samples. Right panel: Validation cohort, n = 350 initial samples. Raw data quality control (QC) metrics were applied after each sample's RNA was assayed on the nCounter Banff Human Organ Transplant panel. Samples passing QC metrics had a housekeeping gene scaling factor < 10 and >62% of probes above the detection threshold. ABMR, antibody-mediated rejection; TCMR, T cell–mediated rejection.

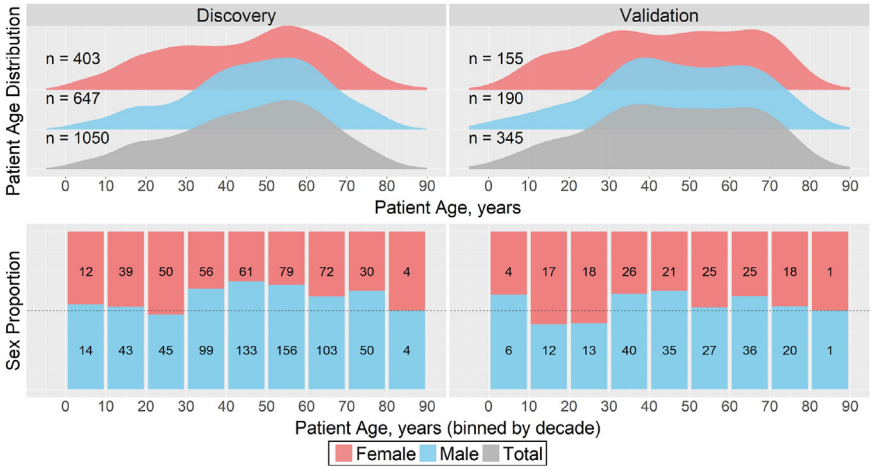

Figure 2.
Patient demographics of samples for model development and validation. Left panels: Discovery cohort. Right panels: Validation cohort. Top panels: Density plots indicating the distribution of patient age at the time of biopsy for all patients (gray), for male patients (blue), and female patients (pink). Lower panels: Bar charts indicating the proportion of male and female patients at each decade of life at the time of biopsy. Numbers within each bar indicate the number of samples for each sex at each decade of life.

cohort (3.2%) and 5 samples in the validation cohort (1.4%) and were excluded from further data analysis (Supplementary Fig. S1). Therefore, the final discovery cohort for model development included 1050 samples representing a diverse range of histologic features binned into the 4 histologic classes (276 ABMR [A_H], 342 TCMR [T_H], 76 mixed rejection [M_H], and 356 no rejection samples (NV_H). The final validation cohort comprised an additional 345 samples (56 A_H, 95 T_H, 18 M_H, and 176 NV_H). Because 2 different RNA extraction protocols were used for samples included in both the final discovery and validation cohorts, principal component analysis (PCA) was performed with the normalized gene expression data of all samples in both cohorts and indicated no systematic variation between samples extracted by the 2 different protocols (Supplementary Fig. S2). PCA was also performed for normalized gene expression data from samples above and those below the median sample age in both discovery (734 days) and validation (498 days) cohorts, and no systematic difference was detected between the 2 groups in either cohort (Supplementary Fig. S2).

Patient Demographics and Sample Characteristics

As shown in Figure 2 and Table 1, study samples represented kidney transplant recipients across the life span (discovery, 3-88 years; mean, 46.1; SD, 17.9; validation, 1-85 years; mean, 45.8; SD, 19.2), with a slight male predominance (discovery, 61.6%; validation, 55.1%). The geographic distribution of sample origin was diverse (Fig. 3), representing 10 of the 11 United Network of Organ Sharing (UNOS) regions, with ≥95 samples from each UNOS region except for regions 1 and 9, and with the greatest representation from region 3 with 358 samples. Of note, ~5% of region 6 cases were from Alaska or Hawaii, and there were 7 international cases. Banff lesion scores of all allograft biopsies included in the discovery and validation cohorts are summarized per histologic class in Table 2, and results highlight the consistency of diagnostic Banff lesion scores within each histologic group between discovery and validation cohorts.

Differential Gene Expression Analysis in the Discovery Cohort

Because a recently published renal allograft biopsy GEP study using the B-HOT gene panel on the nCounter platform successfully validated findings in the extensive existing data generated on microarray platform,[20] we also performed DGE analysis with samples in the discovery cohort to identify the ABMR- and TCMR-associated genes using the similar sample grouping and gene-ranking strategies to the published microarray studies (Fig. 4).[32,33] The top 200 genes associated with either ABMR or TCMR in our study cohort (ranked by false discovery rate [FDR]-adjusted *P* value) were identified and annotated with well-known gene lists from the published microarray studies

Table 1
Summary statistics by histologic group and cohort indicating patient and sample characteristics

Group	Cohort	Patient age, y	Sample age, d	Proportion male, %	No. of cases
NV_H	Discovery	56 (24)	464 (259)	60.1	356
NV_H	Validation	51.5 (29)	431 (128)	58.5	176
A_H	Discovery	50 (23)	1160 (1268)	61.2	276
A_H	Validation	42.5 (25)	714 (1879)	60.7	56
T_H	Discovery	43 (25)	1480 (1617)	62.0	342
T_H	Validation	41 (32)	685 (1662)	44.2	95
M_H	Discovery	45 (23)	1423 (1467)	68.4	76
M_H	Validation	37.5 (10)	1103 (565)	61.1	18

Values for patient age and sample age represent the median, with interquartile range in parentheses.
A_H, antibody-mediated rejection (histologic class); M_H, mixed rejection (histologic class); NV_H, no rejection (histologic class, includes polyomavirus nephritis); T_H, T-cell–mediated rejection (histologic class).

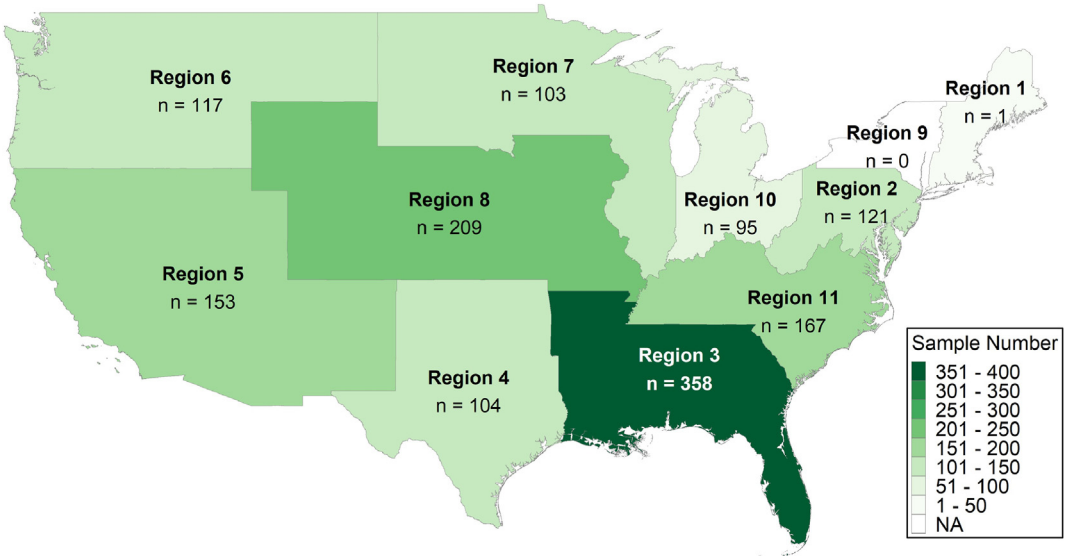

Figure 3.
Geographic distribution of samples by United Network of Organ Sharing (UNOS) region. Samples from both discovery and validation cohorts were mapped by UNOS region based on location of referring center. Samples from Alaska and Hawaii, not depicted, are included in the total number for Region 6. Not shown are 7 international samples (Caribbean and Middle East).

(Supplementary Tables S1 and S2), including "DSA selective transcript" (DSAST) genes,[34] "endothelial cell-associated transcript" (ENDAT) genes,[35,36] "ABMR rejection–associated transcript" (ABMR-RAT) genes and "TCMR rejection–associated transcript" (TCMR-RAT) genes.[32] Of the genes from these 4 lists that are included in the B-HOT panel, nearly all (19 DSAST, 26 ENDAT, 73 ABMR-RAT, 66 TCMR-RAT genes) showed significant DGE (by FDR-adjusted *P* value) when we compared our B-HOT panel data for either ABMR or TCMR samples with all others. Exceptions included only 5 ENDAT genes and 4 ABMR-RAT genes. All 19 DSAST, 20 ENDAT, and 52 ABMR-RAT genes were found among our top 200 ABMR genes, and 64 TCMR-RAT genes were found in our top 200 TCMR genes. Previously published studies reported high similarity of gene expression between TCMR or acute cellular rejection and PVN in renal allograft biopsies.[37,38] Because the NV_H group included biopsies with a histologic diagnosis of PVN, we performed DGE analysis for 66 TCMR-RAT genes included in the B-HOT panel between all 342 TCMR biopsies from the T_H histologic class and all 41 allograft PVN biopsies from the NV_H histologic class in the discovery cohort. Among all 66 TCMR-RAT genes, 63 genes showed significantly higher expression (FDR-adjusted $P < 0.05$) in TCMR than allograft PVN biopsies (Supplementary Table S3), which is very different from the findings in the previous studies.

Multiclass Molecular Classification Model Development and Technical Validation

Using LASSO regularized regression for variable selection, combined with 10-fold cross-validation, we fitted a multiclass model of 4 molecular archetypes/classes (no rejection, including polyomavirus nephritis, NV_M; ABMR, A_M; TCMR, T_M; and mixed rejection, M_M) to the 4 histologic diagnostic classes (NV_H, A_H, T_H, and M_H) using B-HOT panel gene expression data from the 1050 samples in the final discovery cohort. Among the 758 non-housekeeping genes in the B-HOT panel, 143 genes were weighted in the multiclass model, with 20 of these genes weighted across multiple classes (Table 3). Weighted genes in the model cover various cellular and molecular mechanisms contributing to the pathogenesis of rejection, such as renal tissue injury (*NPHS2*, *KAAG1*, *COL4A4*, and *EHD3*), endothelial injury (*ROBO4*, *GNG11*, *PECAM1*, and *NOS3*), inflammatory cytokines/chemokines (*IL18*, *CXCL11*, and *CCL3*), as well as various types of immune cells and

Table 2
Median Banff lesion scores per histologic class in discovery and validation cohorts

Group	Cohort	g	ptc	t	i	c4d	ci	ct	cg	ptcml	v	cv	i-IFTA	t-IFTA	ti
NV_H	Discovery	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (1)	0 (1)	0 (0)	0 (0)	0 (0)	0 (1)	0 (1)	0 (0)	0 (0)
NV_H	Validation	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (1)	0 (1)	0 (0)	0 (0)	0 (0)	0 (1)	0 (1)	0 (1)	0 (1)
A_H	Discovery	3 (2)	2 (1)	0 (1)	0 (1)	2 (3)	1 (1)	1 (1)	1 (3)	0 (0)	0 (0)	1 (2)	1 (1)	1 (1)	1 (1)
A_H	Validation	2 (2)	2 (1)	0 (1)	0 (0)	3 (1.5)	1 (1.25)	1 (1)	0 (2)	0 (0)	0 (0)	1.5 (2.25)	1 (1)	0 (1)	1 (1)
T_H	Discovery	0 (0)	0.5 (1)	3 (1)	2 (1)	0 (0)	1 (1)	1 (1)	0 (0)	0 (0)	0 (0)	0 (1.25)	1 (3)	1 (2)	3 (1)
T_H	Validation	0 (0)	1 (2)	3 (1)	3 (1)	0 (0)	1 (2)	1 (1)	0 (0)	0 (0)	0 (1)	1 (2)	2 (3)	2 (3)	3 (1)
M_H	Discovery	3 (1)	3 (1)	2 (1)	2 (1)	3 (0)	1 (1)	1.5 (1)	1 (2)	0 (1)	0 (1)	1 (2.25)	2 (2)	2 (1)	3 (1)
M_H	Validation	2 (0.75)	2 (1)	2.5 (1)	2.5 (1)	3 (0)	1 (1)	1 (1)	0 (1.75)	0 (1)	1 (1)	2 (1)	1 (0.75)	1 (1.75)	3 (1)

Native kidney biopsies are not included in this table, as they did not undergo Banff lesion scoring.
A_H, antibody-mediated rejection (histologic class); M_H, mixed rejection (histologic class); NV_H, no rejection (histologic class); T_H, T cell–mediated rejection (histologic class). Interquartile range is indicated in parentheses.

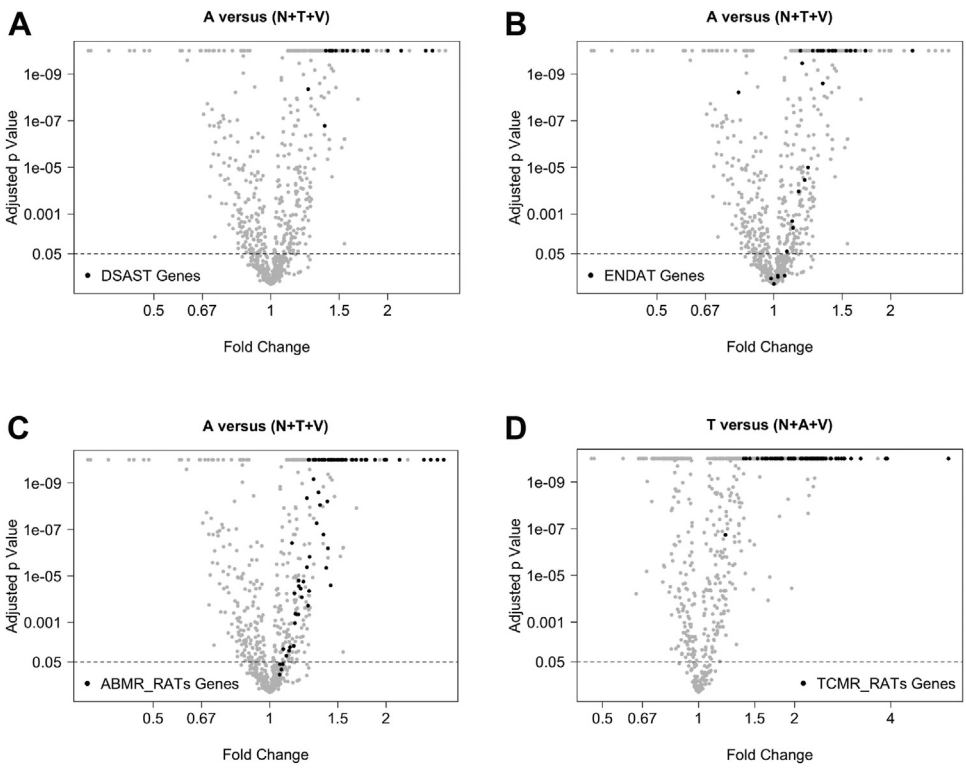

Figure 4.
DGE analysis of all 758 B-HOT panel endogenous genes in the discovery cohort with subsets of genes annotated. Fold change and FDR-adjusted *P* value of each gene in binary comparison of A_H class vs combined $NV_H + T_H$ class (A–C) or T_H class vs combined $NV_H + A_H$ class (D) in the discovery cohort. Black markers indicate genes that are included in the DSAST gene list (A), ENDAT gene list (B), ABMR-RAT gene list (C), or TCMR-RAT gene list (D). ABMR-RAT, ABMR rejection–associated transcript; DSAST, donor-specific antibody–selective transcript; ENDAT, endothelial cell-associated transcript; TCMR-RAT, TCMR rejection–associated transcript.

their activation markers (*CD8B*, *CD80*, *CTLA4*, *SH2D1B*, *CD160*, *IGHG2*, and *FCER1A*).[1,3,39-42]

Accuracy of the model was defined as concurrence of the molecular class with histologic class ("M=H"). In the final discovery cohort, overall accuracy was 84.6%. Sensitivity and specificity of the model to classify "rejection," regardless of histologic type, were 93.4% and 90.4%, respectively. Sensitivity for ABMR and TCMR were each >85%, notably with equal performance in detection of C4d-positive or C4d-negative ABMR (Table 4). Sensitivity for mixed rejection was lower (~49%).

The fitted multiclass molecular model was then applied to an independent final validation cohort of 345 samples. Overall model accuracy was 80.0%, and sensitivity and specificity of the model to classify rejection, regardless of histologic type, were 85.2% and 88.1%, respectively. As shown in Table 4, sensitivity for ABMR was 80.4%, with slightly greater sensitivity for C4d-negative ABMR (92.3%) than for C4d-positive ABMR (76.7%). Sensitivity for TCMR was 71.6% and for mixed rejection was 44.4%.

Discrepancy Analysis Between Histologic and Molecular Diagnosis

Examining the performance of our molecular classification model, we found the highest discrepancy rate between histologic and molecular classes for the Mixed Rejection biopsies (M_H) (Tables 5 and 6). We examined model behavior by plotting the distribution of probabilities of each molecular class for all samples of a given histologic class. As seen in Figure 5, among all M_H samples, the distribution of probabilities for No Rejection (NV_M) appears similar to that seen in the other 3 histologic classes. However, each of the molecular classes of rejection (A_M, T_M, and M_M) shows a more even distribution of probabilities in M_H than in the other 3 histologic classes. These results suggest that although the multiclass model reliably detects rejection in M_H samples, there is model uncertainty in molecular classification of the type of rejection, which may reflect underlying molecular heterogeneity among cases of histologic Mixed Rejection.

When the model was evaluated in the validation cohort, a discrepancy between histologic and molecular class was identified among subgroups of TCMR (T_H) biopsies that lacked a vascular component of rejection, with 30% of those samples classified as NV_M archetype by the model (Table 6). Meanwhile, in the NV_H class, the highest discrepancy rate was found in the allograft acute pyelonephritis and allograft PVN subgroups, with 20% to 25% of the samples in either subgroup assigned a T_M archetype by the model (Table 6). To investigate potential associations of histologic features to class discrepancies, we analyzed Banff lesion scores for the above-mentioned histologic subgroups in the validation cohort. We found that for T_H samples lacking a vascular component of rejection, biopsies discrepantly classified as NV_M by the model had significantly different Banff lesion scores for i, ti, t, and peritubular capillaritis lesion scores (median score 1 grade lower), and ci (median score 1 grade higher), than those classified as T_M (Fig. 6A). However, no significant difference in Banff lesion scores was found between samples in the allograft acute pyelonephritis or allograft PVN subgroups when the molecular class was NV_M compared with T_M (Fig. 6B, C), indicating no significant association of the inflammation burden for the discrepant classification of T_M in these cases.

Table 3
Relative weights of genes included in the multiclass molecular model

Gene name	B-HOT gene panel annotation	Relative gene weight per molecular class			
		NV_M	A_M	T_M	M_M
KIR_inhibiting_subgroup_1	Immune system	100.0	0	0	0
KLRK1	Immune system	84.5	0	0	0
IL7R	Immune system	81.3	0	0	0
BK large T Ag	Viral infection	81.2	0	0	0
HLA-F	Immune system	72.2	0	0	0
PLA1A	Tissue and cellular process	71.7	61.6	61.6	89.3
LGALS3	Immune system	67.5	0	0	0
SH2D1B	Immune system	55.6	47.8	32.0	32.0
SMAD3	Tissue and cellular process	51.9	0	0	0
HLA-C	Immune system	51.2	0	0	0
CXCL11	Immune system	46.2	0	0	0
GBP4	Tissue and cellular process	44.9	0	0	0
SOST	Tissue and cellular process	44.7	7.5	0	0
AGT	Tissue and cellular process	41.3	0	0	0
HSPA12B	Tissue and cellular process	33.5	0	0	0
CD69	Immune system	30.2	0	0	0
ITGA4	Tissue and cellular process	29.5	0	0	0
NCAM1	Immune system	29.5	0	0	0
LCN2	Tissue and cellular process	29.2	0	0	0
ARG2	Tissue and cellular process	28.6	0	0	0
BK VP1	Viral infection	27.1	0	0	0
COL4A1	Tissue and cellular process	25.4	0	0	0
CD59	Immune system	24.1	0	0	0
MCM6	Tissue and cellular process	22.2	0	0	0
XCL1/2	Immune system	21.1	0	0	0
SMARCA4	Tissue and cellular process	18.8	0	0	0
SERPINE1	Tissue and cellular process	17.5	0	0	0
IL18	Immune system	17.0	0	0	0
HLA-DPB1	Immune system	17.0	0	0	0
SOD2	Tissue and cellular process	16.1	0	0	0
CCR7	Immune system	15.9	0	0	0
SIGIRR	Immune system	15.2	0	0	0
CD160	Immune system	13.8	0	0	0
KIT	Tissue and cellular process	13.5	0	0	0
CMV UL83	Viral infection	12.4	0	0	0
TFRC	Tissue and cellular process	10.9	0	0	0
SERINC5	Immune system	9.0	0	0	0
IFIT1	Tissue and cellular process	8.5	0	0	0
HLA-DQA1	Immune system	7.8	0	0	0
IFI6	Tissue and cellular process	7.1	0	0	0
AGR3	Tissue and cellular process	6.6	0	0	0
CXCL8	Immune system	5.9	0	88.2	0
AQP2	Organ specific	5.5	0	16.0	0
MAPK12	Tissue and cellular process	5.0	0	0	0
BCL2L1	Tissue and cellular process	4.8	0	0	0
ROBO4	Tissue and cellular process	4.6	65.8	14.3	4.6
HLA-B	Immune system	4.6	0	0	0
MYBL1	Tissue and cellular process	4.5	0	0	0
EOMES	Immune system	4.3	0	0	0
CFB	Immune system	3.8	0.6	0	0
KLRF1	Immune system	3.4	14.2	0	0
NCR1	Immune system	2.9	0	0	0
LOX	Tissue and cellular process	2.6	0	0	0
GDF15	Tissue and cellular process	2.5	0	17.7	0
CFH	Immune system	1.7	0	2.6	0
CCL21	Immune system	1.6	0	0	0
HFE	Immune system	1.1	0	0	0
CXCR6	Immune system	0	71.7	47.2	0
CXCL13	Immune system	0	69.0	0	0
FCER1A	Immune system	0	67.4	0	0
BTG2	Tissue and cellular process	0	58.7	0	0

(*continued on next page*)

Table 3 (*continued*)

Gene name	B-HOT gene panel annotation	Relative gene weight per molecular class			
		NV_M	A_M	T_M	M_M
CTLA4	Immune system	0	53.7	0	0
CASP3	Tissue and cellular process	0	50.4	0	0
SPRY4	Tissue and cellular process	0	25.5	0	0
RAF1	Tissue and cellular process	0	23.4	0	0
MAPK13	Tissue and cellular process	0	22.3	2.8	0
IGF2R	Tissue and cellular process	0	19.8	0	0
RHOU	Tissue and cellular process	0	17.8	40.8	0
CCL18	Immune system	0	16.4	0	0
LYVE1	Tissue and cellular process	0	15.1	0	0
IL1RL1	Immune system	0	14.5	0	0
CD80	Immune system	0	14.3	1.8	0
KAAG1	Organ specific	0	13.5	0	0
EHD3	Tissue and cellular process	0	12.3	0	0
FOXP3	Tissue and cellular process	0	10.1	0	0
CRIP2	Tissue and cellular process	0	8.8	0	0
TNFSF9	Immune system	0	8.6	0	0
CDH5	Tissue and cellular process	0	8.1	0	0
CD8B	Immune system	0	6.9	0	0
RASSF9	Tissue and cellular process	0	6.6	15.2	0
SIRPG	Tissue and cellular process	0	6.6	0	0
ADORA2A	Tissue and cellular process	0	6.1	0	0
COL4A4	Tissue and cellular process	0	5.0	0	0
PRDM1	Tissue and cellular process	0	4.4	0	0
TRAF4	Immune system	0	4.2	0	0
SOX7	Tissue and cellular process	0	3.6	0	0
JUN	Viral infection	0	3.3	0	0
IL1R2	Immune system	0	3.3	0	19.1
PIN1	Tissue and cellular process	0	2.6	0	0
THBS1	Tissue and cellular process	0	1.5	0	0
RAMP3	Tissue and cellular process	0	1.2	1.7	0
AIRE	Immune system	0	0.9	0	0
GNG11	Tissue and cellular process	0	0	95.2	0
RAPGEF5	Tissue and cellular process	0	0	68.1	0
DEFB1	Immune system	0	0	65.1	0
GNLY	Immune system	0	0	62.8	0
PHEX	Tissue and cellular process	0	0	40.6	0
BMP7	Tissue and cellular process	0	0	34.4	25.3
ENG	Tissue and cellular process	0	0	34.0	0
RELA	Tissue and cellular process	0	0	33.7	0
COL1A1	Tissue and cellular process	0	0	30.7	0
PLAAT4	Tissue and cellular process	0	0	29.2	0
CD40LG	Immune system	0	0	25.2	0
CD81	Tissue and cellular process	0	0	23.4	0
ICAM2	Immune system	0	0	23.0	0
IL33	Immune system	0	0	21.0	0
NPHS2	Organ specific	0	0	20.6	0
CD58	Immune system	0	0	20.3	0
PLAT	Tissue and cellular process	0	0	19.6	0
C5	Immune system	0	0	18.6	0
PECAM1	Immune system	0	0	18.4	0
CD2	Immune system	0	0	17.0	0
EGFR	Tissue and cellular process	0	0	17.0	0
TIPARP	Tissue and cellular process	0	0	16.9	0
TNC	Tissue and cellular process	0	0	16.8	0
MYB	Tissue and cellular process	0	0	15.3	0
BMP2	Tissue and cellular process	0	0	14.8	0
CTNNB1	Tissue and cellular process	0	0	14.5	0
CRHBP	Tissue and cellular process	0	0	11.6	0.6
SLC19A3	Tissue and cellular process	0	0	10.8	0
MT2A	Tissue and cellular process	0	0	9.2	0
EEF1A1	Tissue and cellular process	0	0	9.0	0
BCL2	Tissue and cellular process	0	0	7.8	0
VMP1	Tissue and cellular process	0	0	7.7	0

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Table 3 (*continued*)

Gene name	B-HOT gene panel annotation	Relative gene weight per molecular class			
		NV_M	A_M	T_M	M_M
PSEN1	Tissue and cellular process	0	0	3.9	19.2
TNFSF4	Immune system	0	0	2.4	0
MAPK3	Tissue and cellular process	0	0	1.4	0
TFF3	Tissue and cellular process	0	0	1.3	0
ABCA1	Tissue and cellular process	0	0	1.0	0
MMP9	Tissue and cellular process	0	0	0.8	0
CD55	Immune system	0	0	0.3	0
IGHG2	Immune system	0	0	0.0	93.1
CXCL12	Immune system	0	0	0.0	59.1
CD207	Tissue and cellular process	0	0	0.0	20.4
MICA	Immune system	0	0	0.0	17.0
EPO	Tissue and cellular process	0	0	0.0	16.0
NOS3	Tissue and cellular process	0	0	0.0	13.8
KLF2	Tissue and cellular process	0	0	0.0	9.0
KLF4	Tissue and cellular process	0	0	0.0	7.0
SLC4A1	Tissue and cellular process	0	0	0.0	5.6
CCL3/L1	Immune system	0	0	0.0	4.5
P2RX4	Tissue and cellular process	0	0	0.0	4.4
HPRT1	Tissue and cellular process	0	0	0.0	3.1
Total no. of genes weighted in each molecular class		57	41	51	21

A_M, antibody-mediated rejection (molecular class); M_M, mixed rejection (molecular class); NV_M, no rejection (molecular class, includes polyomavirus nephritis); T_M, T-cell–mediated rejection (molecular class);

Impact of Patient Demographics and Sample Age on Model Performance

Next, to assess the applicability of our model, we evaluated whether patient demographics or sample age impacted model performance. We found no significant difference in median patient age between discovery and validation cohorts ($P = .83$) or between male and female patients ($P = .06$). A significant difference ($P = .04$) in median patient age was found between classification agreement (48.5 years) and classification disagreement (45.0 years). However, mean agreement was generally high across ages (83.8% ± 2.9% at 95% CI), suggesting that our model is robust with respect to sex and across the lifespan (Fig. 7). Median sample age was found to be significantly different ($P < 2 \times 10^{-16}$) between histologic classes, with the median sample age in days as follows: NV_H 456, A_H 1080, T_H 1298, and M_H 1238. However, sample age did not significantly impact model performance (Fig. 8), as median sample age was not significantly different ($P = 0.59$) between samples showing classification agreement (638 days) vs disagreement (778 days).

Discussion

Aided by a very large sample size, diverse histologic findings, and stringent criteria for sample inclusion, we established a robust, GEP-based, multiclass classification model for defining molecular archetypes of renal allograft rejection using data from the B-HOT panel derived from routine diagnostic FFPE biopsies. Among the limitations of our study is the limited amount of paired clinical data, including graft outcome data for the archived samples. Paired data would be useful for a thorough evaluation of the diagnostic accuracy and clinical utility of our model, particularly where discrepancies occur between the molecular and histologic diagnoses. Biopsies were selected on the basis of histologic features, without regard to paired clinical data, by search of consecutive cases from a high-volume national reference laboratory. Therefore, it is possible that some protocol biopsies may have been within the NV histologic group. Likewise, some protocol biopsies may have been within each of the other histologic groups showing rejection, as subclinical rejection does occur. Most protocol biopsies in the United States are performed at large academic transplant centers treating highly

Table 4
Sensitivity and positive predictive value of molecular class for histologic class

	Discovery cohort (%)	Validation cohort (%)
No rejection		
Sensitivity	90.4 ± 1.6	88.1 ± 2.4
PPV	87.5 ± 1.7	86.1 ± 2.6
ABMR		
Sensitivity	85.9 ± 2.1 (C4d+: 85.0 ± 2.6, C4d-: 87.6 ± 3.5)	80.4 ± 5.3 (C4d+: 76.7 ± 6.4, C4d-: 92.3 ± 7.4)
PPV	82.9 ± 2.2	76.3 ± 5.5
TCMR		
Sensitivity	85.4 ± 1.9	71.6 ± 4.6
PPV	83.4 ± 2.0	74.7 ± 4.6
Mixed rejection		
Sensitivity	48.7 ± 5.7	44.4 ± 11.7
PPV	80.4 ± 5.8	53.3 ± 12.9

AMBR, antibody-mediated rejection; PPV, positive predictive value; TCMR, T-cell–mediated rejection.

Table 5
Discovery cohort: Histologic class composition by histologic subgroup with molecular classification results

Histologic class	Histologic subgroup		Total cases, n	QC pass, n	NV_M (%)	A_M (%)	T_M (%)	M_M (%)
NV_H	S_01: Normal (allograft)		40	37	33 (89.2)	2 (5.4)	2 (5.4)	0 (0)
NV_H	S_02: Acute tubular Injury		58	57	52 (91.2)	0 (0)	5 (8.8)	0 (0)
NV_H	S_03: Calcineurin inhibitor toxicity		6	6	5 (83.3)	0 (0)	1 (16.7)	0 (0)
NV_H	S_04: Delayed graft function		20	18	16 (88.9)	1 (5.6)	1 (5.6)	0 (0)
NV_H	S_05: Normal (native kidney)		34	32	32 (100)	0 (0)	0 (0)	0 (0)
NV_H	S_06: Thrombotic microangiopathy (native kidney)		14	14	14 (100)	0 (0)	0 (0)	0 (0)
NV_H	S_07: Acute pyelonephritis (native kidney)		50	50	45 (90.0)	0 (0)	5 (10.0)	0 (0)
NV_H	S_08: CMV nephritis (allograft)		7	6	3 (50.0)	1 (16.7)	2 (33.3)	0 (0)
NV_H	S_09: CMV nephritis (native kidney)		3	3	1 (33.3)	0 (0)	2 (66.7)	0 (0)
NV_H	S_11: Reactive inflammation within areas of fibrosis (native kidney)		44	42	41 (97.6)	1 (2.4)	0 (0)	0 (0)
NV_H	S_12: Diabetic nephropathy (native kidney)		30	28	28 (100)	0 (0)	0 (0)	0 (0)
NV_H	S_13: Smoking-related nodular glomerulosclerosis (native kidney)		16	16	16 (100)	0 (0)	0 (0)	0 (0)
NV_H	S_14: Polyomavirus nephritis (allograft)		44	41	30 (73.2)	2 (4.9)	9 (22.0)	0 (0)
NV_H	S_15: Polyomavirus nephritis (native kidney)		7	6	6 (100)	0 (0)	0 (0)	0 (0)
		No rejection, total	373	356	322 (90.4)	7 (2.0)	27 (7.6)	0 (0)
A_H	S_21: Active ABMR (C4d+)		84	83	7 (8.4)	62 (74.7)	13 (15.7)	1 (1.2)
A_H	S_22: Chronic active ABMR (C4d+)		105	104	1 (1.0)	97 (93.3)	2 (1.9)	4 (3.8)
A_H	S_23: Active ABMR (C4d− DSA+)		41	41	4 (9.8)	33 (80.5)	4 (9.8)	0 (0)
A_H	S_24: Chronic active ABMR (C4d− DSA+)		49	48	3 (6.3)	45 (93.7)	0 (0)	0 (0)
		ABMR, total	279	276	15 (5.4)	237 (85.9)	19 (6.9)	5 (1.8)
T_H	S_16: TCMR 1A		152	145	21 (14.5)	10 (6.9)	114 (78.6)	0 (0)
T_H	S_17: TCMR 1B		152	148	8 (5.4)	4 (2.7)	134 (90.5)	2 (1.4)
T_H	S_18: TCMR 2A		38	35	1 (2.9)	1 (2.9)	33 (94.2)	0 (0)
T_H	S_19: TCMR 2B		8	8	0 (0)	1 (12.5)	6 (75.0)	1 (12.5)
T_H	S_20: TCMR 3		6	6	0 (0)	0 (0)	5 (83.3)	1 (16.7)
		TCMR, total	356	342	30 (8.8)	16 (4.7)	292 (85.4)	4 (1.1)
M_H	S_25: Mixed rejection	Mixed rejection, total	77	76	1 (1.3)	26 (34.2)	12 (15.8)	37 (48.7)

ABMR, antibody-mediated rejection; A_H, antibody-mediated rejection (histologic class); A_M, antibody-mediated rejection (molecular class); M_H, mixed rejection (histologic class); M_M, mixed rejection (molecular class); NV_H, no rejection (histologic class, includes polyomavirus nephritis); NV_M, no rejection (molecular class, includes polyomavirus nephritis); T_H, T-cell−mediated rejection (histologic class); T_M, T-cell−mediated rejection (molecular class); TCMR, T-cell−mediated rejection.

Table 6
Validation cohort: Histologic class composition by histologic subgroup with molecular classification results

Histologic class	Histologic subgroup		Total cases, n	QC pass, n	NV_M (%)	A_M (%)	T_M (%)	M_M (%)
NV_H	S_01: Normal (allograft)		75	74	67 (90.5)	2 (2.7)	5 (6.8)	0 (0)
NV_H	S_02: Acute tubular injury		24	24	24 (100)	0 (0)	0 (0)	0 (0)
NV_H	S_03: Calcineurin inhibitor toxicity		5	5	4 (80.0)	1 (20.0)	0 (0)	0 (0)
NV_H	S_04: Delayed graft function		6	6	6 (100)	0 (0)	0 (0)	0 (0)
NV_H	S_07: Acute pyelonephritis (native kidney)		11	11	11 (100)	0 (0)	0 (0)	0 (0)
NV_H	S_07b: Acute pyelonephritis (allograft)		23	23	17 (73.9)	0 (0)	5 (21.7)	1 (4.3)
NV_H	S_12b: Diabetic nephropathy (allograft)		13	13	13 (100)	0 (0)	0 (0)	0 (0)
NV_H	S_14: Polyomavirus nephritis (allograft)		20	20	13 (65.0)	2 (10.0)	5 (25.0)	0
		No rejection, total	177	176	155 (88.1)	5 (2.8)	15 (8.5)	1 (0.6)
A_H	S_21: Active ABMR (C4d+)		33	32	4 (12.5)	23 (71.9)	4 (12.5)	1 (3.1)
A_H	S_22: Chronic active ABMR (C4d+)		11	11	0 (0)	10 (90.9)	0 (0)	1 (9.1)
A_H	S_23: Active ABMR (C4d- DSA+)		2	2	0 (0)	2 (100)	0 (0)	0 (0)
A_H	S_24: Chronic active ABMR (C4d- DSA+)		11	11	1 (9.1)	10 (90.9)	0 (0)	0 (0)
		ABMR, total	57	56	5 (8.9)	45 (80.4)	4 (7.1)	2 (3.6)
T_H	S_16: TCMR 1A		30	28	12 (42.9)	2 (7.1)	14 (50.0)	0 (0)
T_H	S_17: TCMR 1B		36	35	7 (20.0)	1 (2.9)	23 (65.7)	4 (11.4)
T_H	S_18: TCMR 2A		21	21	0 (0)	1 (4.8)	20 (95.2)	0 (0)
T_H	S_19: TCMR 2B		9	9	0 (0)	0 (0)	9 (100)	0 (0)
T_H	S_20: TCMR 3		2	2	0 (0)	0 (0)	2 (100)	0 (0)
		TCMR, total	98	95	19 (20.0)	4 (4.2)	68 (71.6)	4 (4.2)
M_H	S_25: Mixed rejection	Mixed rejection, total	18	18	1 (5.6)	5 (27.8)	4 (22.2)	8 (44.4)

ABMR, antibody-mediated rejection; A_M, antibody-mediated rejection (molecular class); A_H, antibody-mediated rejection (histologic class); M_H, mixed rejection (histologic class); M_M, mixed rejection (molecular class); NV_H, no rejection (histologic class, includes polyomavirus nephritis); T_H, T-cell−mediated rejection (histologic class); NV_M, no rejection (molecular class, includes polyomavirus nephritis); T_M, T-cell−mediated rejection (molecular class); TCMR, T-cell−mediated rejection.

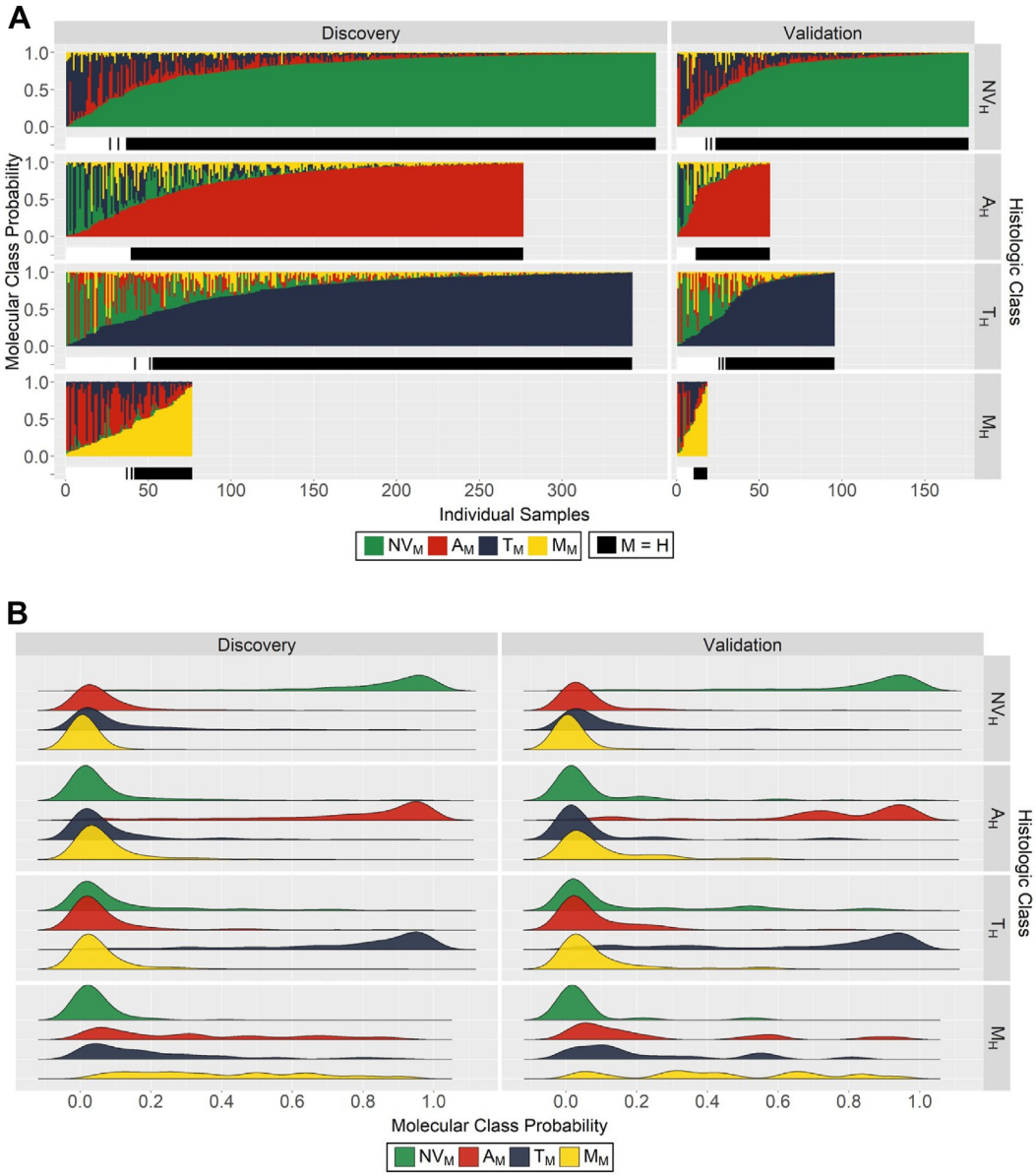

Figure 5.
Individual sample molecular class probability scores and their distribution within each histologic class. (A) Molecular class probability scores of each sample. Left panels: Discovery cohort. Right panels: Validation cohort. Stacked bar charts, grouped by histologic class, indicate the molecular class probability scores generated by the model for each sample of that histologic class. Beneath the stacked bar chart for each histologic class is a black bar indicating samples with matched molecular and histologic diagnoses (M = H). (B) Distribution of molecular class probability scores. Left panels: Discovery cohort. Right panels: Validation cohort. Density plots of probabilities for each molecular class determined by the multiclass model among samples of a given histologic class.

sensitized patients, and these large academic transplant centers often have their own in-house renal pathology diagnostic service. Given that our reference laboratory serves most of the remaining transplant centers and private practices in the United States, which lack in-house renal pathology services, it is reasonable to assume that our biopsy cohort did not contain a significant proportion of protocol biopsies.

Furthermore, to refine our approach in generating the first multiclass model from a large cohort using the B-HOT panel, and due to ongoing debate in the field related to true disease state of some diagnostic categories, we did not include in our cohorts cases diagnosed as Borderline (Suspicious) for Acute TCMR, Chronic (Inactive) ABMR, or Chronic Active TCMR having only minimal or mild interstitial inflammation (i < 2). Future studies will evaluate these diagnostic categories, and application of our model to these types of cases may shed light on the underlying molecular pathogenesis of these contested diagnostic categories. Even better, pairing patients' comprehensive clinical profiles and long-term graft outcome data to stratify fraught diagnostic categories prior to dissection by our molecular model may be of even greater value for the field and for future studies defining the clinical utility of our approach.

To train the multiclass model, we carefully created a cohort with clear histologic findings of TCMR, ABMR, and Mixed Rejection. To better simulate the range of possible non-rejection diagnoses in the renal biopsy setting, we included non-rejection

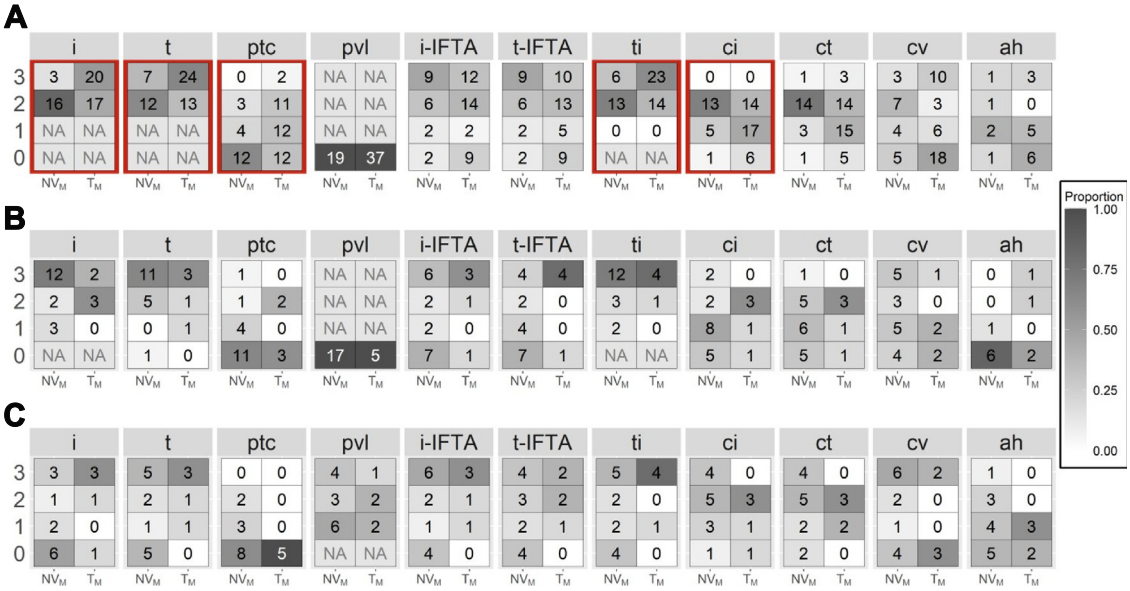

Figure 6.

Banff lesion scores of samples from the validation cohort showing discordance of molecular and histologic class. The numbers (0, 1, 2, 3) to the left of each panel indicate the Banff lesion scores. The numbers in each cell of each box indicate the case count of biopsies with that Banff lesion score. Boxes with red outlines indicate those Banff lesions for which scores were significantly different between molecular classes ($P < .05$) by the Cochrane-Armitage test for Trend. Gray shaded cells with "NA" indicate Banff lesion scores which by diagnostic definition are not possible in particular histologic groups. (A) Tubulointerstitial-type TCMR (Banff grades 1A and 1B) samples from histologic class T_H with model-assigned molecular class of either NV_M or T_M. (B) Acute pyelonephritis (allograft) samples from histologic class NV_H with model-assigned molecular class of either NV_M or T_M. (C) Polyomavirus nephritis (allograft) samples from histologic class NV_H with model-assigned molecular class of either NV_M or T_M. NV_H, no rejection, histologic class, includes polyomavirus nephritis; NV_M, no rejection, molecular class, includes polyomavirus nephritis; T_H, TCMR, histologic class; T_M, TCMR, molecular class.

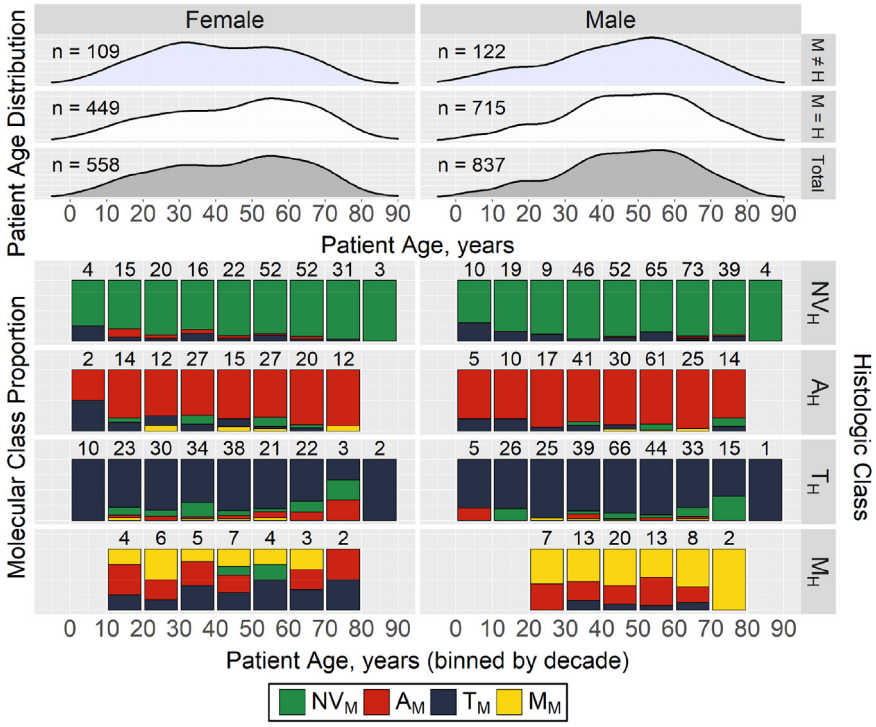

Figure 7.

Patient age distribution by gender and effect on model performance. Left panels: Female. Right panels: Male. Top panel: Density plots of patient age distribution. Top row: Patient age distribution for cases with mismatched molecular and histologic class (pale blue). Middle row: Patient age distribution for cases with matched molecular and histologic class (white). Bottom row: Patient age distribution across all cases (gray). Bottom panel: Stacked bar charts grouped by histologic class, showing proportion of each molecular class by patient age as binned by decade. Numbers at the top of each stacked bar indicate total number of cases within each patient age bin for that histologic class. Results indicate analysis of combined final discovery and validation cohorts.

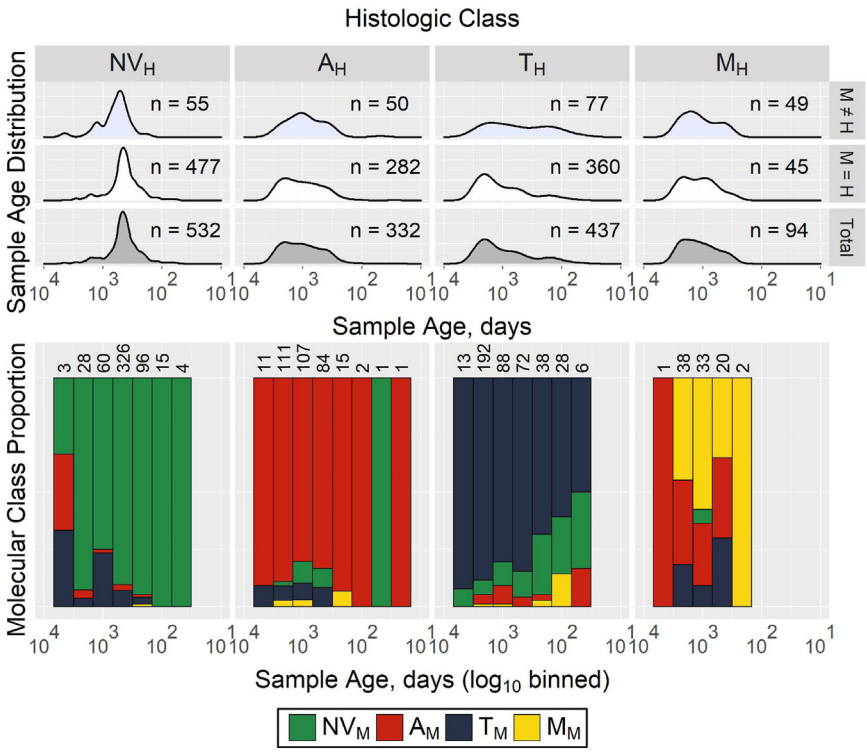

Figure 8.
Sample age distribution and effect on model performance. Top panel: Density plots of sample age distribution per histologic class. Top row: Sample age distribution for cases of mismatched molecular and histologic class (pale blue). Middle row: Sample age distribution for cases of matched molecular and histologic class (white). Bottom row: Sample age distribution across all cases per histologic class (gray). Bottom panel: Stacked bar charts grouped by histologic class, showing proportion of each molecular class by sample age as binned by $\log_{10}$ days. Numbers at the top of each stacked bar indicate total number of cases within each sample age bin for that histologic class.

biopsy specimens with very diverse histologic findings, including biopsies with inflammation attributed to other causes. One limitation of this study is that, while we did include cases of mild glomerular IgA deposition in the allograft, we did not include examples of recurrent glomerulonephritis, a very interesting though relatively rare occurrence, which could be a focus for future studies in order to assess molecular differences/similarities between glomerulonephritis and allograft rejection. Another limitation was that urine culture results were not always available in cases of clear-cut histologic acute pyelonephritis. When the diagnostic model was applied to the validation cohort, it successfully classified many of the transplant biopsies with inflammation unrelated to rejection in the "No Rejection" class, including all the diabetic nephropathy cases and the majority of the acute pyelonephritis and PVN cases (Table 6).

It is worth clarifying our logic for constructing the "No Rejection" histologic class for the discovery cohort. We included subgroups of samples with various histologic diagnosis/findings not attributable to allograft rejection, which may seem counterintuitive or perhaps incorrect at first glance. These subgroups included samples of PVN cases and acute pyelonephritis in allografts as well as in native kidneys. The cellular infiltrate in these types of cases can pose challenges in the histologic evaluation of allograft biopsies when ruling out rejection is a main concern. Therefore, we intentionally included samples with cellular infiltrates not attributable to allograft rejection in the discovery cohort in order to focus the training of the molecular class (T_M) on the core molecular archetype of TCMR, and remove from T_M as much influence from histologic mimics.

Other groups have sought to separate by molecular means No Rejection, PVN, and TCMR. However, we view of primary importance whether or not the biopsy diagnosis triggers further immunosuppression. Therefore, in order to avoid detrimental further immunosuppression in cases of PVN or acute pyelonephritis, we weighted the "No Rejection" histologic class with examples of these cases. We believe that our molecular model is successful to this end because it maintains a good concurrence with definitive histologic ABMR and TCMR in the validation cohort, while classifying all the transplant diabetic nephropathy cases and the majority of transplant acute pyelonephritits and PVN cases in the validation cohort as No Rejection (NV_M).

Published studies focusing on GEP in PVN biopsies, from either native kidney or renal allografts, have been relatively limited with the majority of them reporting similarity of gene expression between TCMR and PVN.[37,38,43,44] In our study, of the roughly 25% of cases of PVN cases misclassified by our model, TCMR was also the most often assigned molecular class (Tables 5 and 6). However, when we performed DGE analysis for the well-published and validated TCMR-associated genes (TCMR-RAT; 66 of which are included in the B-HOT panel) in cases of histologic TCMR and allograft PVN, we found that almost all these TCMR-RAT genes (63 of 66) showed significantly higher expression in cases of histologic TCMR than in allograft PVN biopsies in our discovery cohort (Supplementary Table S3). These results clearly indicate a significant difference in overall intragraft gene expression between histologic TCMR and allograft PVN in our study cohort, and the overlap of a minority of histologic PVN cases with the molecular archetype of TCMR suggests heterogeneity of gene expression within allograft PVN cases. Meanwhile, ~30% of samples in the T_H class of the validation cohort were classified by our model to be non-TCMR. These "misclassified" TCMR samples were

predominantly of the tubulointerstitial-type (Banff grade IA and IB) and were most often classified as NV_M by our model. Analysis of Banff lesion scores in these cases and the subgroups of allograft PVN or allograft acute pyelonephritis in the validation cohort found an association with the degree of inflammatory cell infiltrate and misclassification as NV_M only in cases of TCMR, but no significant differences in Banff lesion scores were found between concordant and discordant calls in pyelonephritis or PVN. Although the numbers in these specific scenarios are small, and only associations can be determined from these observations, future studies could be designed to investigate how the degree or character of the cellular infiltrate associates with the molecular archetype. Indeed, future clinical validation studies with paired clinical outcomes data, treatment regimen data, and so on may shed more light on etiologies of discordant histologic diagnosis and molecular archetypes.

Overall, the lowest concordance rate between histologic and molecular classes was found in the M_H histologic class. However, it is worth noting that the accuracy of our model for confirming histologic Mixed Rejection was not inferior to that of a commercially available microarray-based diagnostic system.[45] With rare exceptions, "misclassified" M_H samples were predicted to be either A_M or T_M. This result is perhaps not surprising, as Mixed Rejection was defined as the presence of morphologic features ascribed to both ABMR and TCMR. That the molecular model misclassified some mixed rejection cases in this manner may simply reflect an underlying and/or perhaps momentary molecular leaning within a heterogenous disease process exhibiting the morphologic hallmarks of both. Again, future studies with follow-up clinical data may better inform the significance of these molecular "misclassifications."

Historically, many GEP studies of renal allograft biopsy tissue have focused on identifying genes associated with either a specific type of rejection or a specific type of Banff histologic lesion using a binary comparison strategy, then from these results built gene classifiers or molecular models for the diagnosis of rejection or other pathologic conditions. [12,32,39,46,47] Given the thorough selection criteria for genes included in the B-HOT panel,[9] we opted to use all assayed endogenous genes in the B-HOT panel (except 13 genes with expression specific to non-kidney solid organs) as input features for model fitting instead of prespecifying features by any binary comparison. Due to the statistical nature of LASSO model fitting,[30,31,48] the genes and their specific weighting in the molecular classifier of our diagnostic model for each diagnostic class do not necessarily agree perfectly with the conventional understanding of those genes' association with any particular type of renal allograft rejection (often inferred by simple binary comparison of one rejection type with the other types of rejection or nonrejection disease conditions). Although, as a testament to the veracity of the many transplant molecular studies that supported the development of the B-HOT panel, many of the genes included in our model were previously reported to be highly associated with either ABMR, TCMR, or rejection in general, such as *PLA1A*, *SH2D1B*, *ROBO4*, *CXCR6*, *CTLA4*, *CXCL11*, and *XCL1*. We also performed DGE analysis in the discovery cohort by binary comparison using similar sample grouping and gene-ranking strategies to the published microarray studies and identified ABMR- or TCMR-associated genes in our study. We compared our DGE analysis results with the well-published renal allograft rejection-related gene lists established by the microarray studies and validated those gene lists in our study cohort successfully, which supports the validity of GEP findings in our study.

It is also worth noting that our study was not designed as a clinical validation of our novel classification model nor as a guide to the development of a clinical test that can be run in a reference laboratory. Our intention and achievement in this study was to develop and validate a classification model based on transcriptional data derived from clinically relevant material and clinically feasible tools (archived diagnostic FFPE tissue and the B-HOT gene expression panel) where each class in our model represents the molecular "archetype" of a histologic class, and where samples have well-curated histology and numerous examples of the range of that histology. Future studies are required and are underway to test the clinical validity and utility of assigning a molecular archetype from data derived from the B-HOT GEP panel, as well as to address the real-world issues of maintaining assay validation prospectively over time, which will have to include methods to account for variation between successive batch runs.

Of note, 2 different lots of B-HOT gene panel assays were used in this study. PCA showed a very minimal impact of this lot change on raw gene expression data, with lot-associated sample grouping by raw gene expression data observed only in principal component 5, which accounted for <3% of total variability. Applying the ComBat-seq procedure mitigated this minimal batch effect on the raw gene expression data. Nonetheless, removing the ComBat-seq procedure resulted in no significant change in model performance, with a molecular class change seen in <4% of the validation cohort samples, and the overall model performance accuracy remained almost identical. Future studies employing a greater number of lots of B-HOT gene panel assays may be needed to assess the true impact of assay lot change on raw gene expression data and model performance.

In summary, with a very large sample cohort of diverse histologic diagnoses, low-QC failure rates from samples up to 13 years old with no detectable relationship between sample age and QC failure, our study confirmed the technical feasibility and unique advantages of performing retrospective allograft biopsy GEP studies on the nCounter platform. In doing so, we created a multiclass model for assigning molecular archetypes of rejection, with results showing high correlation between histologic and molecular classes of rejection. Ongoing and future studies, some of which are illustrated above, will be necessary to define clinical utility of these molecular archetypes. Indeed, a remaining question for the field, in general, is in which clinical scenarios adjunct information from molecular testing of graft tissue, such as a molecular archetype, could positively impact patient care. Generation of the robust model described here was the necessary first step to move the well-designed B-HOT gene panel, which generates data, toward the common future goal of clinically feasible molecular tests having defined clinical utility to guide patient care.

Author Contributions

H.Z., F.C., N.A., G.S., K.Q., R.W., S.S., R.H., C.C., J.N., J.W., C.L., and S.C. performed study concept and design. H.Z., F.C., N.A., G.S., T.S., K.Q., R.W., S.S., R.H., C.C., J.N., J.W., S.H., C.L., and S.C. performed development of methodology and writing and review and revision of the paper. H.Z., F.C., W.T., K.Q., R.H., C.C., J.N., J.W., S.H., K.A., C.B., N.M., T.C., L.C., S.S., J.A., C.L., and S.C. provided acquisition, analysis, and interpretation of data, and statistical analysis. The contributions by N.A., G.S., T.S., K.Q., and S.S. were made while they were employees of CareDx. The contributions of K.A. and N.M. were made while they were employees of Arkana Laboratories. All authors read and approved the final paper.

Data Availability

Upon publication, the raw data that support the findings of this study will be openly available in the NCBI Gene Expression Omnibus (GEO) at https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE212160

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Declaration of Competing Interest

H. Zhang, F. Collin, G. Shekhtman, W. Tian, T. Srinivas, K. Qu, R.N. Woodward are salaried employees of CareDx; N. Agrawal and S. Stone were salaried employees of CareDx; N. Messias was a salaried employee of Arkana Laboratories. R.S. Haun, C. Cassol, J. Wilson, J.O.H. Napier, C. Boils, T.N. Caza, L.N. Cossey, S. Sharma, J.M. Ambruzs, S. Hassen, C.P. Larsen, and S.M. Coley are salaried employees of Arkana Laboratories. N. Agrawal owns stocks of AstraZeneca. CareDx applied for patents related to the content of this manuscript. K. Ararat reports no conflict of interest.

Ethics Approval and Consent to Participate

The study protocol was approved by the Solutions Institutional Review Board, and this study was performed in accordance with the Declaration of Helsinki principles.

Supplementary Material

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